

STA 5635 HW 7

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Problem 1

Batch size = 128.

(1a) NN w/ one hidden layer, 256 neurons, ReLU activation.

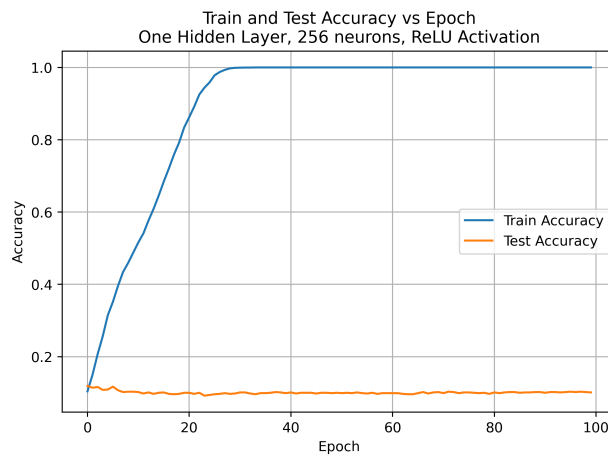


Figure 1: Train and Test accuracy vs. Epoch, part a.

(1b) NN w/ two hidden layers, 64 neurons each, ReLU activation

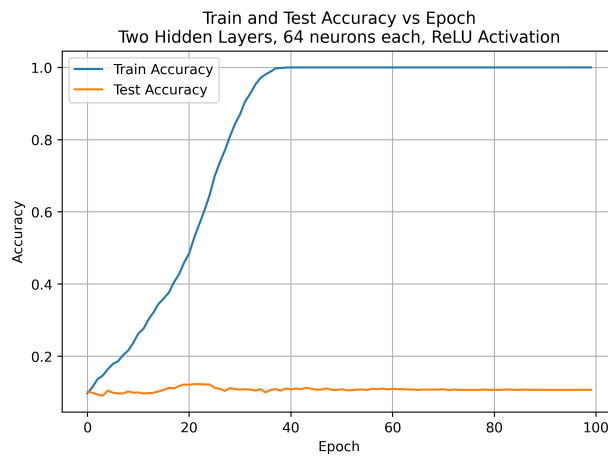


Figure 2: Train and Test accuracy vs. Epoch, part b.

(1c) CNN w/ one hidden layer, 32 filters of size 15, ReLU activation.

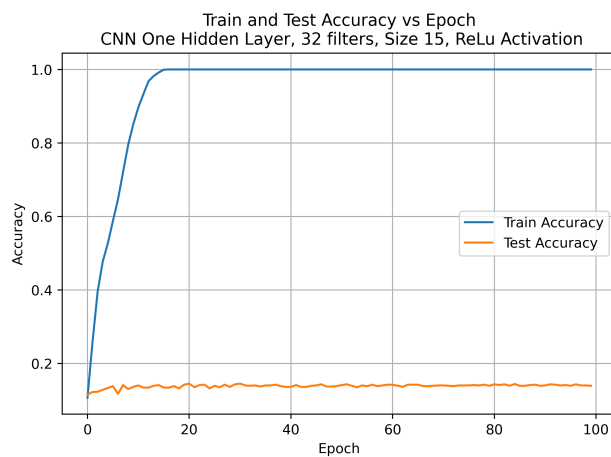


Figure 3: Train and Test accuracy vs. Epoch, part c.

(1d) CNN w/ one hidden layer, 32 filters of size 15, ReLU activation, max pooling size 2, stride 2

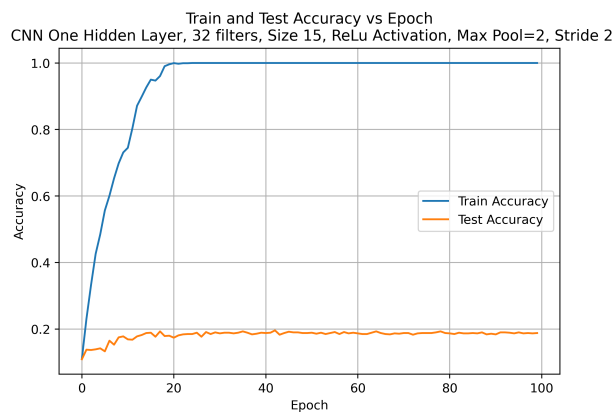


Figure 4: Train and Test accuracy vs. Epoch, part d.

- (1e) CNN w/ one hidden layer, 32 filters of size 15, ReLUE activation, max pooling entire output.

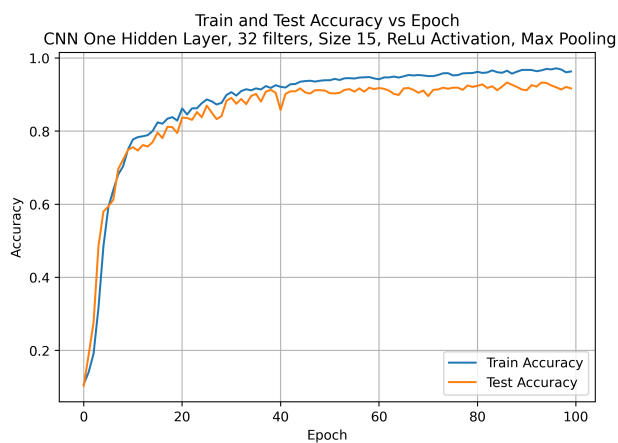


Figure 5: Train and Test accuracy vs. Epoch, part e.

- (1f) CNN w/ two hidden layers, 16 filters of size 15, ReLUE activation, max pooling size 2, stride 2.

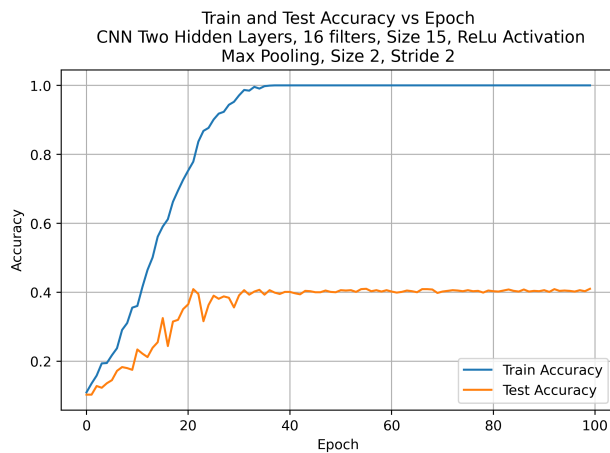


Figure 6: Train and Test accuracy vs. Epoch, part f.

- (1g) CNN w/ three hidden layers, 16 filters of size 15, ReLUE activation, max pooling size 2, stride 2.

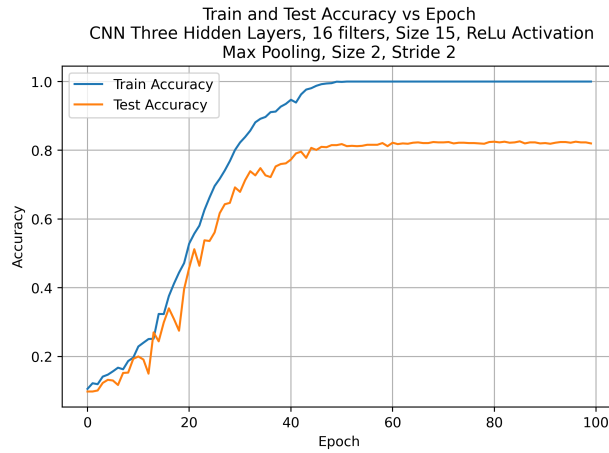


Figure 7: Train and Test accuracy vs. Epoch, part g.

- (1h) Table of final train and test accuracies. From this, it appears as though the model in

model	final_train_acc	final_test_acc
part_a	1.000000	0.101000
part_b	1.000000	0.107000
part_c	1.000000	0.139000
part_d	1.000000	0.188000
part_e	0.963333	0.917000
part_f	1.000000	0.410000
part_g	1.000000	0.820000

Table 1: Final Train and Test Accuracies for all Models

part e had the best test accuracy.